

Conducting Forensic Investigations on Linux Systems (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

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Section 1: Hands-On Demonstration

Part 1: Explore a Live Linux System

17. Make a screen capture showing the contents of the /bin directory.

A terminal window titled 'Conducting Forensic Investigations on Linux Systems (4e)' showing the command `ls -l /bin` and its output. The output lists various system binaries with their permissions, owner, group, size, and date. The prompt is `user@TargetLinux01:/bin$`.

```
ls -l /bin
lrwxrwxrwx 1 root root 10 Apr 20 2020 xzgrep -> xzgrep
lrwxrwxrwx 1 root root 10 Apr 20 2020 xzfgrep -> xzfgrep
-rwxr-xr-x 1 root root 1028 Apr 20 2020 xzgrep
-rwxr-xr-x 1 root root 1802 Apr 20 2020 xzless
-rwxr-xr-x 1 root root 2161 Apr 20 2020 xzmore
-rwxr-xr-x 1 root root 63720 Apr 7 2020 yelp
-rwxr-xr-x 1 root root 39256 Sep 5 2019 yes
lrwxrwxrwx 1 root root 10 May 12 2021 ypdomainname -> hostname
-rwxr-xr-x 1 root root 1984 Dec 13 2019 zcat
-rwxr-xr-x 1 root root 1678 Dec 13 2019 zcmp
-rwxr-xr-x 1 root root 5880 Dec 13 2019 zdiff
-rwxr-xr-x 1 root root 26840 Dec 16 2020 zdump
-rwxr-xr-x 1 root root 29 Dec 13 2019 zegrep
-rwxr-xr-x 1 root root 135960 Feb 27 2020 zenity
-rwxr-xr-x 1 root root 29 Dec 13 2019 zfgrep
-rwxr-xr-x 1 root root 2081 Dec 13 2019 zforce
-rwxr-xr-x 1 root root 7585 Dec 13 2019 zgrep
-rwxr-xr-x 1 root root 216256 Apr 21 2017 zip
-rwxr-xr-x 1 root root 93816 Apr 21 2017 zipcloak
-rwxr-xr-x 1 root root 50718 Oct 19 2020 zipdetails
-rwxr-xr-x 1 root root 2953 Aug 16 2019 zipgrep
-rwxr-xr-x 1 root root 186664 Aug 16 2019 zipinfo
-rwxr-xr-x 1 root root 89488 Apr 21 2017 zipnote
-rwxr-xr-x 1 root root 93584 Apr 21 2017 zipsplit
-rwxr-xr-x 1 root root 26952 Jan 30 2020 zjsdecode
-rwxr-xr-x 1 root root 2206 Dec 13 2019 zless
-rwxr-xr-x 1 root root 1842 Dec 13 2019 zmore
-rwxr-xr-x 1 root root 4553 Dec 13 2019 znew
```

20. Make a screen capture showing the contents of the /etc directory.

A terminal window titled 'Conducting Forensic Investigations on Linux Systems (4e)' showing the command `ls -l /etc` and its output. The output lists various system configuration files with their permissions, owner, group, size, and date. The prompt is `user@TargetLinux01:/etc$`.

```
ls -l /etc
drwxr-xr-x 2 root root 4096 May 17 2021 .
drwxr-xr-x 5 root root 4096 May 17 2021 ..
drwxr-xr-x 2 root root 4096 Apr 23 2020 .terminfo
drwxr-xr-x 2 root root 4096 May 17 2021 thermal
-rw-r--r-- 1 root root 17 May 17 2021 timezone
drwxr-xr-x 2 root root 4096 Apr 22 2020 tmpfiles.d
drwxr-xr-x 2 root root 4096 May 17 2021 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Dec 14 2018 ucf.conf
drwxr-xr-x 4 root root 4096 May 17 2021 udev
drwxr-xr-x 2 root root 4096 Apr 23 2020 udisks2
drwxr-xr-x 3 root root 4096 Apr 23 2020 ufw
drwxr-xr-x 3 root root 4096 May 17 2021 update-manager
drwxr-xr-x 2 root root 4096 May 17 2021 update-motd.d
drwxr-xr-x 2 root root 4096 Apr 2 2020 update-notifier
drwxr-xr-x 2 root root 4096 Apr 23 2020 UPower
-rw-r--r-- 1 root root 1523 Feb 10 2020 usb_modeswitch.conf
drwxr-xr-x 2 root root 4096 Feb 24 2020 usb_modeswitch.d
drwxr-xr-x 2 root root 4096 Apr 23 2020 vim
drwxr-xr-x 4 root root 4096 May 17 2021 vmware-tools
-rw-r--r-- 1 root root 23 May 12 2021 vtrgb -> /etc/alternatives/vtrgb
drwxr-xr-x 5 root root 4096 Apr 23 2020 vulkan
-rw-r--r-- 1 root root 4942 Jul 24 2019 wgetrc
drwxr-xr-x 2 root root 4096 May 17 2021 wpa_supplicant
drwxr-xr-x 11 root root 4096 Apr 23 2020 X11
-rw-r--r-- 1 root root 642 Sep 23 2019 xattr.conf
drwxr-xr-x 6 root root 4096 Apr 23 2020 xdg
drwxr-xr-x 2 root root 4096 Apr 23 2020 xml
-rw-r--r-- 1 root root 477 Oct 7 2019 zsh_command_not_found
```

21. Make a screen capture showing the contents of the /var directory.

```

user@TargetLinux01:~$ cd /var
user@TargetLinux01:/var$ ls -l
total 48
drwxr-xr-x 2 root root 4096 Jan 3 2022 backups
drwxr-xr-x 17 root root 4096 May 17 2021 cache
drwxrwsrwt 2 root whoopsie 4096 Apr 23 2020 crash
drwxr-xr-x 66 root root 4096 Aug 20 2021 lib
drwxrwsr-x 2 root staff 4096 Apr 15 2020 local
lrwxrwxrwx 1 root root 9 May 12 2021 lock -> /run/lock
drwxrwxr-x 13 root syslog 4096 Apr 11 17:03 log
drwxrwsr-x 2 root mail 4096 Apr 23 2020 mail
drwxrwsrwt 2 root whoopsie 4096 Apr 23 2020 metrics
drwxr-xr-x 2 root root 4096 Apr 23 2020 opt
lrwxrwxrwx 1 root root 4 May 12 2021 run -> /run
drwxr-xr-x 12 root root 4096 Jan 3 2022 snap
drwxr-xr-x 7 root root 4096 Aug 20 2021 spool
drwxrwxrwt 10 root root 4096 Apr 11 17:05 tmp

```

22. Make a screen capture showing the contents of the /proc directory.

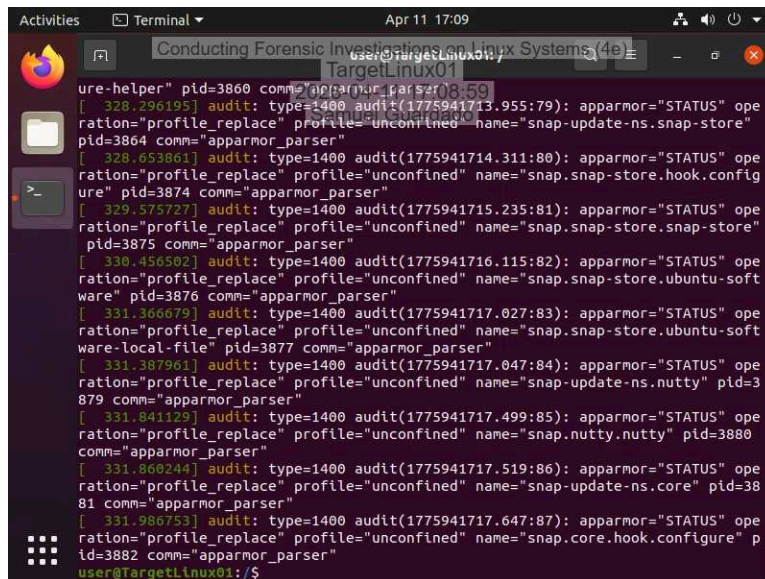
```

user@TargetLinux01:~$ cd /proc
user@TargetLinux01:/proc$ ls -l
-r--r--r-- 1 root 0 Apr 11 17:08 st
at
-r--r--r-- 1 root 0 Apr 11 17:03 sw
aps
dr-xr-xr-x 1 root root 0 Apr 11 17:03 sy
s
--w----- 1 root root 0 Apr 11 17:03 sy
srq-trigger
dr-xr-xr-x 5 root root 0 Apr 11 17:08 sy
svipc
lrwxrwxrwx 1 root root 0 Apr 11 17:03 th
read-self -> 3878/task/3878
-r----- 1 root 0 Apr 11 17:08 ti
mer_list
dr-xr-xr-x 6 root root 0 Apr 11 17:08 tt
y
-r--r--r-- 1 root root 0 Apr 11 17:08 up
time
-r--r--r-- 1 root root 0 Apr 11 17:08 ve
-r--r--r-- 1 root root 0 Apr 11 17:08 ve
rsion_signature
-r----- 1 root root 0 Apr 11 17:08 vn
allocinfo
-r--r--r-- 1 root root 0 Apr 11 17:08 vn
stat
-r--r--r-- 1 root root 0 Apr 11 17:08 zo
neinfo

```

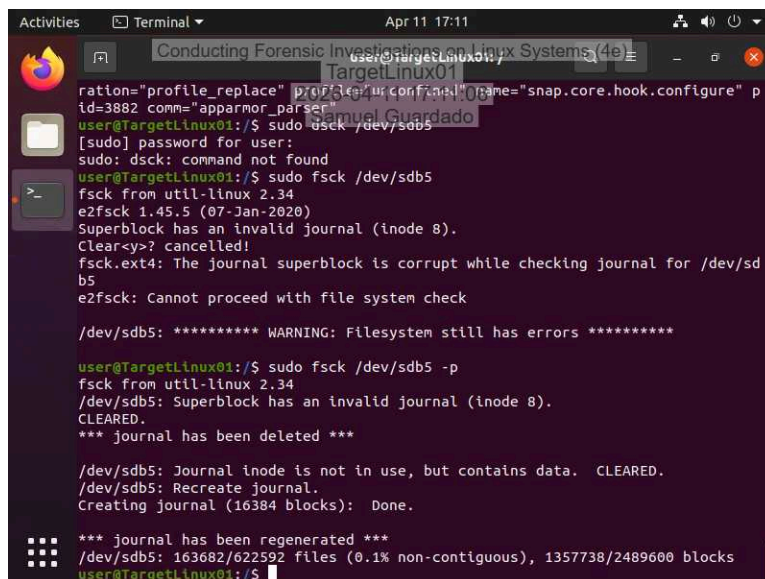
Part 2: Use Linux Shell Commands for Forensic Investigations

2. Make a screen capture showing the results of the dmesg command.

A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing a series of audit logs. The logs are generated by the apparmor_parser process, with various audit messages indicating profile replacements and status changes. The user is identified as "user@TargetLinux01".

```
ure-helper" pid=3860 comm="apparmor_parser"
[ 328.296195] audit: type=1400 audit(1775941713.955:79): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap-update-ns.snap-store"
pid=3864 comm="apparmor_parser"
[ 328.653861] audit: type=1400 audit(1775941714.311:80): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.hook.config
ure" pid=3874 comm="apparmor_parser"
[ 329.575727] audit: type=1400 audit(1775941715.235:81): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.snap-store"
pid=3875 comm="apparmor_parser"
[ 330.456502] audit: type=1400 audit(1775941716.115:82): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-soft
ware" pid=3876 comm="apparmor_parser"
[ 331.366679] audit: type=1400 audit(1775941717.027:83): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-soft
ware-local-file" pid=3877 comm="apparmor_parser"
[ 331.387961] audit: type=1400 audit(1775941717.047:84): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap-update-ns.nutty" pid=3
879 comm="apparmor_parser"
[ 331.841129] audit: type=1400 audit(1775941717.499:85): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.nutty.nutty" pid=3880
comm="apparmor_parser"
[ 331.860244] audit: type=1400 audit(1775941717.519:86): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap-update-ns.core" pid=38
81 comm="apparmor_parser"
[ 331.986753] audit: type=1400 audit(1775941717.647:87): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.core.hook.configure" p
id=3882 comm="apparmor_parser"
user@TargetLinux01:/$
```

7. Make a screen capture showing the results of the fsck command.

A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing the execution of the fsck command. The user is prompted for a password and then runs "sudo fsck /dev/sdb5". The output shows that the superblock has an invalid journal (inode 8) and that the journal is corrupted. The user then runs "sudo fsck /dev/sdb5 -p", which results in the journal being regenerated. The user is identified as "user@TargetLinux01".

```
ration="profile_replace" profile="unconfined" name="snap.core.hook.configure" p
id=3882 comm="apparmor_parser"
user@TargetLinux01:/$ sudo fsck /dev/sdb5
[sudo] password for user:
sudo: dsck: command not found
user@TargetLinux01:/$ sudo fsck /dev/sdb5
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
Superblock has an invalid journal (inode 8).
Clear-y>? cancelled!
fsck.ext4: The journal superblock is corrupt while checking journal for /dev/sd
b5
e2fsck: Cannot proceed with file system check

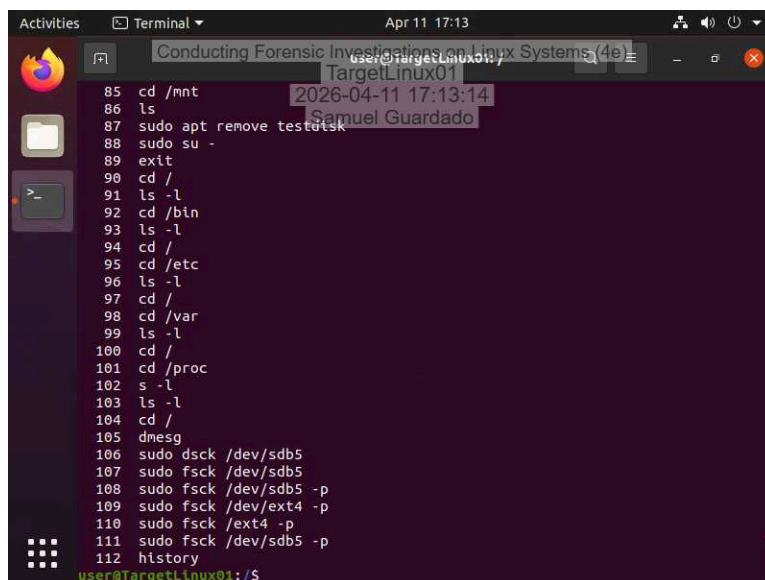
/dev/sdb5: ***** WARNING: Filesystem still has errors *****

user@TargetLinux01:/$ sudo fsck /dev/sdb5 -p
fsck from util-linux 2.34
/dev/sdb5: Superblock has an invalid journal (inode 8).
CLEARED.
*** journal has been deleted ***

/dev/sdb5: Journal inode is not in use, but contains data. CLEARED.
/dev/sdb5: Recreate journal.
Creating journal (16384 blocks): Done.

*** journal has been regenerated ***
/dev/sdb5: 163682/622592 files (0.1% non-contiguous), 1357738/2489600 blocks
user@TargetLinux01:/$
```

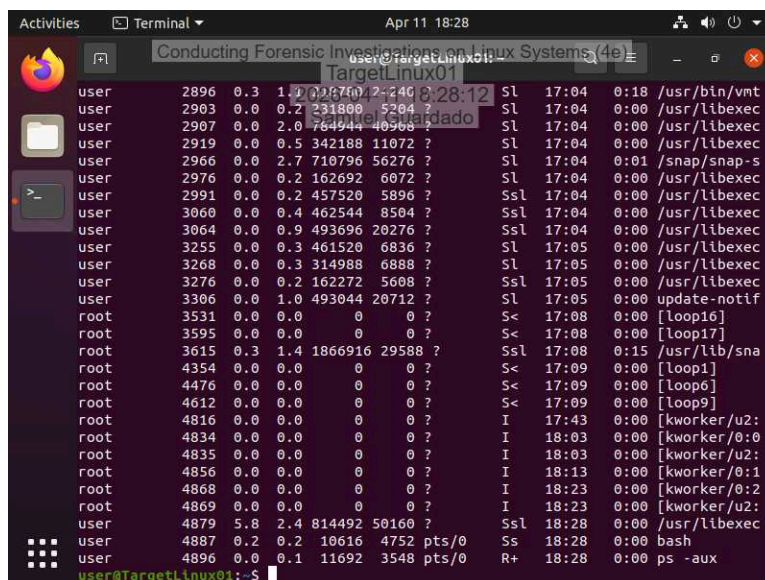
9. Make a screen capture showing the results of the history command.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" with a subtitle "TargetLinux01". The terminal shows a series of commands and their outputs, ending with the 'history' command. The output of 'history' lists 12 commands, including directory navigation, file listing, package removal, and disk checks. The prompt is 'user@TargetLinux01:/\$'.

```
85 cd /mnt
86 ls
87 sudo apt remove testdisk
88 sudo su -
89 exit
90 cd /
91 ls -l
92 cd /bin
93 ls -l
94 cd /
95 cd /etc
96 ls -l
97 cd /
98 cd /var
99 ls -l
100 cd /
101 cd /proc
102 s -l
103 ls -l
104 cd /
105 dmesg
106 sudo dsck /dev/sdb5
107 sudo fsck /dev/sdb5
108 sudo fsck /dev/sdb5 -p
109 sudo fsck /dev/ext4 -p
110 sudo fsck /ext4 -p
111 sudo fsck /dev/sdb5 -p
112 history
user@TargetLinux01:/$
```

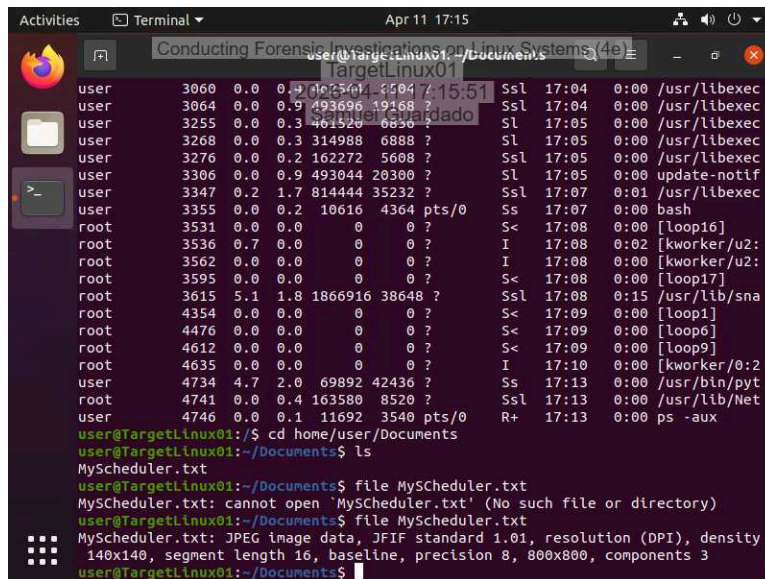
11. Make a screen capture showing the running processes.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" with a subtitle "TargetLinux01". The terminal shows the output of the 'ps -aux' command, displaying a list of running processes with columns for user, PID, CPU, MEM, VSZ, RSS, T, S, PPID, P, W, STIME, USER, and COMMAND. The prompt is 'user@TargetLinux01:~\$'.

```
user      2896  0.3  1.2 201670 12340 ?        Sl   17:04   0:18 /usr/bin/vmt
user     2903  0.0  0.2 231800  5204 ?        Sl   17:04   0:00 /usr/libexec
user     2907  0.0  2.0 764944 40960 ?        Sl   17:04   0:00 /usr/libexec
user     2919  0.0  0.5 342188 11072 ?        Sl   17:04   0:00 /usr/libexec
user     2966  0.0  2.7 710796 56276 ?        Sl   17:04   0:01 /snap/snap-s
user     2976  0.0  0.2 162692  6072 ?        Sl   17:04   0:00 /usr/libexec
user     2991  0.0  0.2 457520  5896 ?        Ssl  17:04   0:00 /usr/libexec
user     3060  0.0  0.4 462544  8504 ?        Ssl  17:04   0:00 /usr/libexec
user     3064  0.0  0.9 493696 20276 ?        Ssl  17:04   0:00 /usr/libexec
user     3255  0.0  0.3 461520  6836 ?        Sl   17:05   0:00 /usr/libexec
user     3268  0.0  0.3 314988  6888 ?        Sl   17:05   0:00 /usr/libexec
user     3276  0.0  0.2 162272  5608 ?        Ssl  17:05   0:00 /usr/libexec
user     3306  0.0  1.0 493044 20712 ?        Sl   17:05   0:00 update-notif
root     3531  0.0  0.0  0  0 ?        S<   17:08   0:00 [loop16]
root     3595  0.0  0.0  0  0 ?        S<   17:08   0:00 [loop17]
root     3615  0.3  1.4 1866916 29588 ?        Ssl  17:08   0:15 /usr/lib/sna
root     4354  0.0  0.0  0  0 ?        S<   17:09   0:00 [loop1]
root     4476  0.0  0.0  0  0 ?        S<   17:09   0:00 [loop6]
root     4612  0.0  0.0  0  0 ?        S<   17:09   0:00 [loop9]
root     4816  0.0  0.0  0  0 ?        I   17:43   0:00 [kworker/u2:
root     4834  0.0  0.0  0  0 ?        I   18:03   0:00 [kworker/0:0
root     4835  0.0  0.0  0  0 ?        I   18:03   0:00 [kworker/u2:
root     4856  0.0  0.0  0  0 ?        I   18:13   0:00 [kworker/0:1
root     4868  0.0  0.0  0  0 ?        I   18:23   0:00 [kworker/0:2
root     4869  0.0  0.0  0  0 ?        I   18:23   0:00 [kworker/u2:
user     4879  5.8  2.4 814492 50160 ?        Ssl  18:28   0:00 /usr/libexec
user     4887  0.2  0.2 10616  4752 pts/0    Ss   18:28   0:00 bash
user     4896  0.0  0.1 11692  3548 pts/0    R+   18:28   0:00 ps -aux
user@TargetLinux01:~$
```

15. Make a screen capture showing the results of the file command.

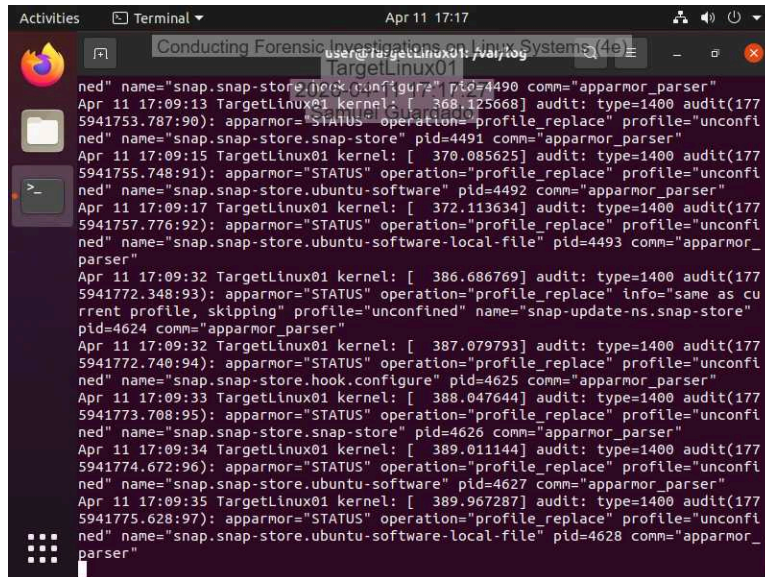


A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing system logs and file operations on a system named "TargetLinux01". The logs display various system events, including network connections, process execution, and file operations. The user "user" is logged in. The terminal shows the following commands and output:

```
user@TargetLinux01:/$ cd /home/user/Documents
user@TargetLinux01:~/Documents$ ls
MyScheduler.txt
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: cannot open 'MyScheduler.txt' (No such file or directory)
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: JPEG image data, JFIF standard 1.01, resolution (DPI), density
140x140, segment length 16, baseline, precision 8, 800x800, components 3
user@TargetLinux01:~/Documents$
```

Part 3: Retrieve Logs Files on a Live Linux System

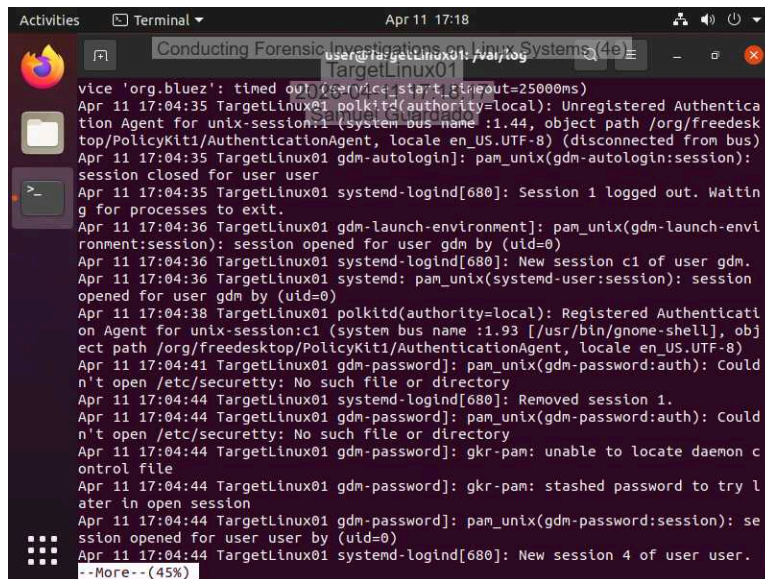
4. Make a screen capture showing the records in the kern.log file.



A terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" showing audit logs in the kern.log file on a system named "TargetLinux01". The logs display various system events, including network connections, process execution, and file operations. The user "user" is logged in. The terminal shows the following commands and output:

```
user@TargetLinux01:/$ cat /var/log/kern.log
Apr 11 17:09:13 TargetLinux01 kernel: [ 368.125668] audit: type=1400 audit(1775941753.787:90): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=4491 comm="apparmor_parser"
Apr 11 17:09:15 TargetLinux01 kernel: [ 370.085625] audit: type=1400 audit(1775941755.748:91): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=4492 comm="apparmor_parser"
Apr 11 17:09:17 TargetLinux01 kernel: [ 372.113634] audit: type=1400 audit(1775941757.776:92): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=4493 comm="apparmor_parser"
Apr 11 17:09:32 TargetLinux01 kernel: [ 386.686769] audit: type=1400 audit(1775941772.348:93): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap-update-ns.snap-store" pid=4624 comm="apparmor_parser"
Apr 11 17:09:32 TargetLinux01 kernel: [ 387.079793] audit: type=1400 audit(1775941772.740:94): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=4625 comm="apparmor_parser"
Apr 11 17:09:33 TargetLinux01 kernel: [ 388.047644] audit: type=1400 audit(1775941773.708:95): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=4626 comm="apparmor_parser"
Apr 11 17:09:34 TargetLinux01 kernel: [ 389.011144] audit: type=1400 audit(1775941774.672:96): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=4627 comm="apparmor_parser"
Apr 11 17:09:35 TargetLinux01 kernel: [ 389.967287] audit: type=1400 audit(1775941775.628:97): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=4628 comm="apparmor_parser"
```

7. Make a screen capture showing the records in the auth.log file.



Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

The two non root users were Dominic, and Noel. With 22 attempts, with the source IP: 192.78.1 on port 2511 on the 11th of June, with a time range of 00:57:19-05:39:01

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

Noel was not able to get into the system, on the other hand, Dominic's last successful login was on June 11th, at 05:23:03.

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

Logkeys, autotools-dev, autoconf, kbd, cacti, and openssh.serverSuSpicious activity: Logkeys. Openssh casn be used for remote access, and possibly gain back door access. Cacti is a network monitoring tool used by attackers to gain unauthorized access to networks, execute remote commands, and steal data.

Part 3: Identify External Drive Attachments on a Linux Drive Image

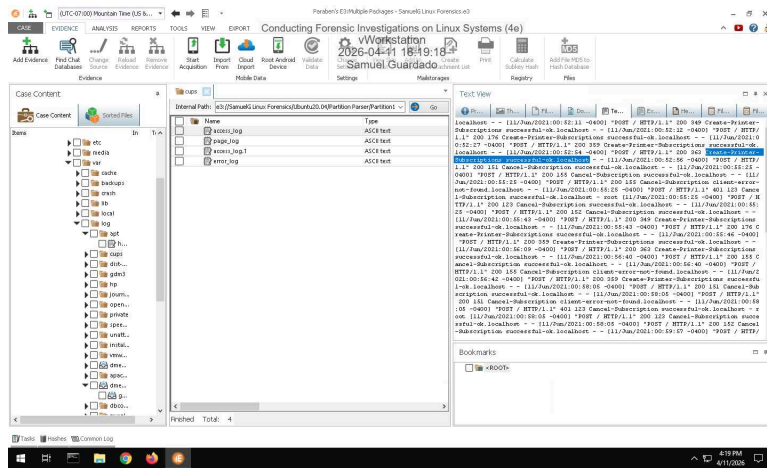
4. **Document** when the USB storage device was connected and its serial number.

It was detected on Jun 10 at 10:24:12. Serial Number: FBI1405291710344

Section 3: Challenge and Analysis

Part 1: Identify Recently Printed Files on a Linux Drive Image

Make a screen capture showing the **contents of the printer log file**.



Part 2: Identify Disk Imaging on a Linux Drive Image

Make a screen capture showing the record of the dd command in the Text View.

